Problem 2: Mars Rover Sample Collection Planning using PDDL

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Files: Domain.pddl, Problem.pddl, Problem_extended.pddl, Problem2_Mars_Rover_PDDL.ipynb. Planner outputs, plans, VAL results and screenshots are in the planner_output folder.

Task 1: Domain Modeling

Requirements: :strips and :typing only. There are no numeric fluents and no negative preconditions.

Types: rover, location, rock, bay, drill.

Predicates: rover-at(rover, location), connected(from, to), rock-at(rock, location), soft(rock), hard(rock), drill-at(drill, location), has-drill(rover), bay-of(bay, rover), empty(bay), in-bay(rock, bay).

Actions:

Action	Preconditions	Effects
move	rover at from, connected(from, to)	rover at to, not at from
pick-drill	rover and drill at same location	has-drill, drill no longer at location
collect-soft	rover and rock at same location, rock is soft, bay belongs to rover and is empty	rock in bay, rock not at location, bay not empty
collect-hard	same as collect-soft, plus rock is hard and rover has the drill	rock in bay, rock not at location, bay not empty
deposit	rover at location, rock in bay	rock at location, bay empty, rock not in bay

Without negative preconditions, an empty bay is tracked with a positive fact, empty(bay). Collecting removes it and depositing adds it back, so a bay can never hold two rocks. The one-way slope is modelled by listing only connected(sitec, sited), with no reverse fact. Soft and hard rocks get separate collect actions, so a hard rock can only be collected once has-drill is true.

Task 2: Problem Modeling

Objects: curiosity (rover); base, sitea to sitef (locations); rock1 to rock5; bay1, bay2; drill1.

Initial state: the rover is at base with both bays empty, and drill1 is at sitee. The rocks start at sitea, siteb, sitec, sited and sitef. Rock1, rock2 and rock4 are soft; rock3 and rock5 are hard. All two-way routes are listed in both directions, plus the single fact connected(sitec, sited).

Goal: all five rocks at base, bay1 and bay2 empty, and the rover at base.

Task 3: Plan Generation

Planner: Fast Downward. Satisficing configuration: greedy best-first search with the FF heuristic, eager_greedy([ff()]). Optimal configuration: A* with the admissible iPDB heuristic, astar(ipdb()). Every action costs 1, so plan length, plan cost and total number of actions are the same number. The full action sequences are in planner_output/*.plan and in the notebook.

Scenario	Configuration	Plan length	Total actions	States expanded	Search time (s)	Total time (s)
Base	Satisficing (GBFS + FF)	30	30	51	0.0010	0.0042
Base	Optimal (A* + iPDB)	25	25	1,930	0.0059	1.1225
Extended	Satisficing (GBFS + FF)	51	51	79	0.0014	0.0036
Extended	Optimal (A* + iPDB)	40	40	868,407	1.9373	4.3092

Optimal plan, base scenario (25 actions):

1 move base→siteb, 2 collect rock2→bay1, 3 move siteb→sited, 4 collect rock4→bay2, 5 move sited→siteb, 6 move siteb→base, 7 deposit rock2, 8 deposit rock4,
9 move base→sitea, 10 move sitea→sitec, 11 move sitec→sitee, 12 pick-drill, 13 move sitee→sitef, 14 collect-hard rock5→bay1, 15 move sitef→sitee, 16 move sitee→sitec,
17 collect-hard rock3→bay2, 18 move sitec→sitea, 19 move sitea→base, 20 deposit rock3, 21 move base→sitea, 22 collect rock1→bay2, 23 move sitea→base, 24 deposit rock1, 25 deposit rock5.

Task 4: Goal Verification

All four plans were checked in two ways. VAL reported "Plan valid" for every plan. The notebook also replays each plan one action at a time, checking the preconditions before applying the effects, and then compares the final state with the goal.

Check	Base (both plans)	Extended (both plans)
Rocks at Base	rock1 to rock5 all at base	rock1 to rock7 all at base
Bay1 / Bay2	empty / empty	empty / empty
Rover location	base	base
Drill	carried by rover	carried by rover
All goal facts satisfied	Yes	Yes

VAL result	Plan valid	Plan valid

Task 5: Extended Scenario

Problem_extended.pddl adds rock6 (soft) at sitee and rock7 (hard) at sitef, and removes both connected facts between base and siteb. The goal is rock1 to rock7 at base with the rover at base. Both configurations found valid plans: 51 actions (satisficing) and 40 actions (optimal).

Optimal extended plan, as trips from Base:

Trip 1: base→sitea→sitec→sited→siteb→sited→sitee→sitec→sitea→base, carrying rock2 and rock4 (drill picked up at sitee on the way back)
Trip 2: base→sitea→sitec→sitee→sitef→sitee→sitec→sitea→base, carrying rock5 and rock7
Trip 3: base→sitea→sitec→sitee→sitec→sitea→base, carrying rock6 and rock3
Trip 4: base→sitea→base, carrying rock1

Task 6: Plan Analysis

6.1 Plan lengths. Base: 30 (satisficing) vs 25 (optimal). Extended: 51 vs 40. Both plans collect and deposit every rock once and pick up the drill once, so the whole difference comes from extra moves. Greedy best-first search only follows the heuristic and ignores how many steps it has already used. So it takes the first decent plan it finds, which here means mostly one rock per trip with bay2 left empty. The base satisficing plan makes 4 trips and the extended one makes 6, against 3 and 4 in the optimal plans. A* with an admissible heuristic must prove that no shorter plan exists, so its plans are as short as possible. The cost of that is search effort: 1,930 vs 51 states in the base case, and 868,407 vs 79 in the extended case.

6.2 Bays and drill (optimal plans). In the base plan, the first trip fills both bays with the soft rocks on the SiteB side (rock2 in bay1, rock4 in bay2). The second trip again fills both bays, with rock5 and rock3. Rock3 is dropped at Base, but rock5 stays in bay1 while the rover makes a quick trip to SiteA and brings rock1 back in bay2. Then both are unloaded. This costs the same number of actions as unloading rock5 earlier. The drill is picked up at step 12, on the second trip, at SiteE just before SiteF. That is the first moment a hard rock is needed, and SiteE is on the way. In the extended plan, every trip uses both bays except the last one, because 7 rocks is an odd number. The drill is picked up at step 9, on the first trip: the rover has to pass through SiteE on its way back from SiteD anyway, so it grabs the drill then.

6.3 Reaching SiteB and SiteD after the dust storm. With Base↔SiteB closed, SiteB can only be reached through SiteD. The optimal plan goes Base→SiteA→SiteC→SiteD on the one-way slope, then SiteD→SiteB. The SiteD→SiteC direction is not allowed, so the rover returns SiteB→SiteD→SiteE→SiteC→SiteA→Base and collects the drill at SiteE. So yes, the one-way route is used, in both the optimal and satisficing extended plans. It saves a step compared with going SiteC→SiteE→SiteD. In the base scenario, the optimal plan does not use the slope, because SiteB and SiteD are reached directly from Base. The base satisficing plan does use it once, on its third trip.